Mammalia Raccoon Proximity Network Analysis

https://github.com/aiyshwary/Raccoon-Network-Analysis

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations thr
centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality
individuals and quantify potential rabies spread pathways within urban populations.

KEY DETAILS

1. Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Pythor
2. Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by t
3. Identified superspreader nodes whose removal would most disrupt disease propagation chain
4. Visualized network topology and centrality distributions to communicate findings clearly.
5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.
6. Analyzed a dense 24-node, 226-edge proximity graph (~82% density) derived from 1,997 weig
7. Computed clustering coefficients and aggregated per-node contact strength (total edge weight
   node 4 had the highest total contact weight (956,341).
8. Rendered the network using Kamada-Kawai force-directed layout (Matplotlib) to reveal spatial
   data.